

Questions

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Q1 - 2024 (04 Apr Shift 1)

The co-ordinates of a particle moving in $x - y$ plane are given by : $x = 2 + 4t, y = 3t + 8t^2$.

The motion of the particle is :

- (1) uniformly accelerated having motion along a parabolic path.
- (2) uniform motion along a straight line.
- (3) uniformly accelerated having motion along a straight line.
- (4) non-uniformly accelerated.

Q2 - 2024 (05 Apr Shift 2)

A man carrying a monkey on his shoulder does cycling smoothly on a circular track of radius 9 m and completes 120 revolutions in 3 minutes. The magnitude of centripetal acceleration of monkey is (in m/s^2) :

- (1) $57600\pi^2 \text{ ms}^{-2}$
- (2) Zero
- (3) $4\pi^2 \text{ ms}^{-2}$
- (4) $16\pi^2 \text{ ms}^{-2}$

Q3 - 2024 (05 Apr Shift 2)

The maximum height reached by a projectile is 64 m. If the initial velocity is halved, the new maximum height of the projectile is _____ m.

Q4 - 2024 (08 Apr Shift 2)

The angle of projection for a projectile to have same horizontal range and maximum height is :

- (1) $\tan^{-1}(4)$
- (2) $\tan^{-1}\left(\frac{1}{4}\right)$
- (3) $\tan^{-1}\left(\frac{1}{2}\right)$
- (4) $\tan^{-1}(2)$

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Q5 - 2024 (08 Apr Shift 2)

A body of mass M thrown horizontally with velocity v from the top of the tower of height H touches the ground at a distance of 100 m from the foot of the tower. A body of mass $2M$ thrown at a velocity $\frac{v}{2}$ from the top of the tower of height $4H$ will touch the ground at a distance of _____ m.

Questions

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Answer Key

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Q5 (100) MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

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Solutions

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Q1

$$x = 2 + 4t$$

$$\frac{dx}{dt} = v_x = 4$$

$$\frac{dv_x}{dt} = a_x = 0$$

$$y = 3t + 8t^2$$

$$\frac{dy}{dt} = v_y = 3 + 16t$$

$$\frac{dv_y}{dt} = a_y = 16$$

the motion will be uniformly accelerated motion.

For path

$$x = 2 + 4t$$

$$\frac{(x-2)}{4} = t$$

Put this value of t is equation of y

$$y = 3\left(\frac{x-2}{4}\right) + 8\left(\frac{x-2}{4}\right)^2$$

this is a quadratic equation so path will be parabola.

Q2

Given : $R = 9 \text{ m}$,

120 revolution in 3 min

$$\omega = \frac{120 \text{ Rev.}}{3 \text{ min.}} = \frac{120 \times 2\pi \text{ rad}}{3 \times 60 \text{ sec}} = \frac{4\pi}{3} \text{ rad/s}$$

$$a_{\text{centripetal}} = \omega^2 R = \left(\frac{4\pi}{3}\right)^2 \times 9 = 16\pi^2 \text{ m/s}^2$$

Q3

$$H_{\max} = \frac{u^2 \sin^2 \theta}{2g}$$

$$\frac{H_{1\max}}{H_{2\max}} = \frac{u_1^2}{u_2^2}$$

$$\frac{64}{H_{2\max}} = \frac{u^2}{(u/2)^2}$$

$$H_{2\max} = 16 \text{ m}$$

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Solutions

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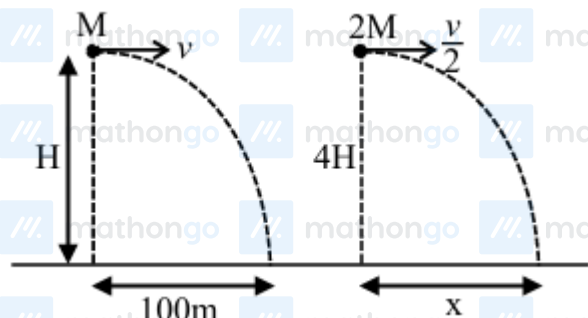
Q4

$$\frac{u^2 \sin 2\theta}{g} = \frac{u^2 \sin^2 \theta}{2g}$$

$$4 \sin \theta \cos \theta = \sin^2 \theta$$

$$4 = \tan \theta$$

Q5



$$100 = v \sqrt{\frac{2H}{g}}; \quad x = \frac{v}{2} \sqrt{\frac{2(4H)}{g}} = v \sqrt{\frac{2H}{g}}$$

$$\Rightarrow x = 100$$

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